

B.3 Consistency

In Appendix Figure 12, we visualize the per-topic alignment of LMs along the fine-grained topics displayed in Appendix Table 3. We construct this figure, as well as Figure 5 as follows. Let’s say we have a model M with a per-question opinion distribution of $D_M(q)$. Further, consider a demographic attribute L (e.g., political ideology) with corresponding subgroups $G_1, G_2, \dots, G_l$ (very liberal, liberal, ..., very conservative). Further, say that the dataset topics are grouped into topic categories $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_K$ (e.g., abortion, personal finance, ...).

For each topic $\mathcal{T}_k$, we consider the dataset questions $Q_{\mathcal{T}_k}$ belonging to that topic. On these questions, we then find the best representative subgroup as:

$$G_{\mathcal{T}_k}^{best} = \arg \max_{G \in \{G_1, G_2, \dots, G_l\}} \mathcal{R}_M^G(Q_{\mathcal{T}_k}) \quad (3)$$

We also assign a significance score to this group as

$$\alpha_{\mathcal{T}_k}^{best} = \frac{\max_{G \in \{G_1, G_2, \dots, G_l\}} \mathcal{R}_M^G(Q_{\mathcal{T}_k})}{\min_{G \in \{G_1, G_2, \dots, G_l\}} \mathcal{R}_M^G(Q_{\mathcal{T}_k})} \quad (4)$$

In Figures 5 and Appendix Figure 12, we then denote the $G_{\mathcal{T}_k}^{best}$ for each topic using a color, and the significance $\alpha_{\mathcal{T}_k}^{best}$ using dot size. For instance, a large red dot implies that a model is strongly aligned with conservatives on that topic.

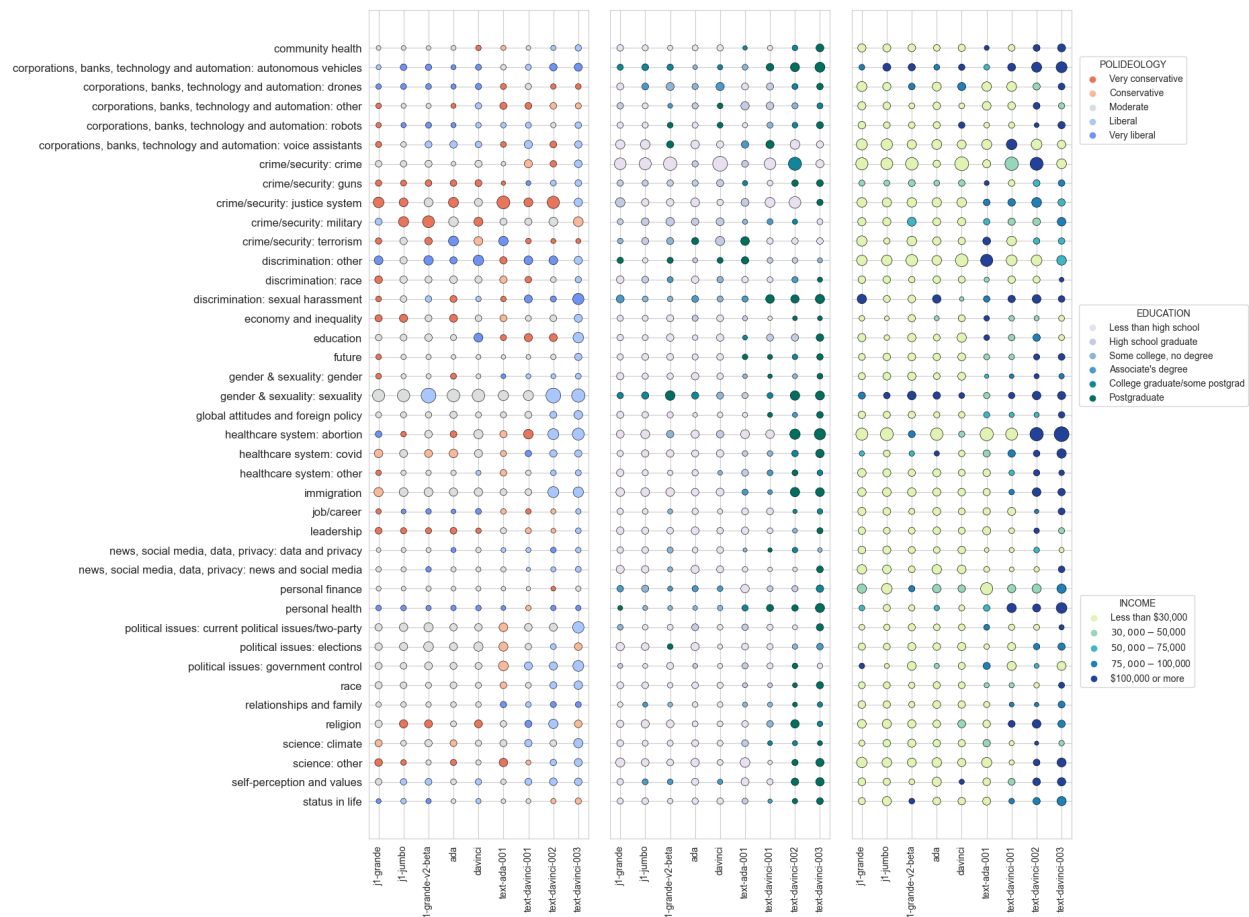


Figure 12: Subgroups that various LMs are best aligned with by fine-grained topic (indicated by dot color), along the axes of political ideology, education, and income levels. The size of the dot indicates how significant the bias towards that group is: computed as the ratio of the best and worst subgroup representativeness for that topic.

B.4 Robustness

Although current LMs perform remarkably well in the zero-shot setting, they are still known to be sensitive to the exact format of their prompt (see Gao et al. (2021); Liang et al. (2022); Srivastava et al. (2022) for extensive evaluations). Thus, one might wonder: Are the distributions we are obtaining from LMs robust to such design choices? Before we delve into this further, it is important to note that humans also exhibit a similar sensitivity. In the context of Pew surveys, human respondents are also sensitive to factors such as option ordering and question formatting. Nevertheless, we test how robust our analysis is to: (i) the order in which options for a question are presented to the model and (ii) prompt formatting. Even though we see small fluctuations in the actual representativeness scores through these interventions, the overall trends remain unchanged—the relative ranking of models and the subgroups they tend to align with.

B.4.1 Sensitivity to option ordering

We exactly repeat our analysis from the main paper, but present the model with answer choices for a question in a randomly permuted (rather than the default ordinal) order. For instance, for the question in Figure 1, we might present the options as “A: Not too much, B: A great deal, C: A fair amount, D: Not at all”. For a given question, the same random permutation is used across LMs.

Under such permutations, we see a small drop in the representativeness scores of all models. We believe that this is at least partly because the reference human distribution is based on survey responses where humans were presented options in an ordinal manner rather than randomly. Since humans are also sensitive to option ordering, we believe this has some effect on the observed human opinion distribution. However, as mentioned above, the overall and subgroup-level trends remain largely consistent as seen from Figure 13.

B.4.2 Sensitivity to prompt format

We vary prompt we feed into LMs so as to get their opinion distribution. Specifically, before asking the model a question—as in Figure 1, we consider adding a set of instructions. The instructions are in one of two formats:

General:

Please read the following multiple-choice question carefully and select ONE of the listed options.

Example:

Please read the multiple-choice question below carefully and select ONE of the listed options. Here is an example of the format:

Question: Question_1

A. Option_1

B. Option_2

C. Option_3

Answer: C

In both cases, the instruction is followed by the question of interest from the dataset.

We then repeat our analysis with these prompt variants (where ‘standard’ denotes our approach from the main paper), focusing on the 500 questions from Section 4.2 computational reasons—see Appendix Figure 14. We only include a subset of demographic attributes in the figure below for brevity, as the results are similar to Appendix Figure 13.